Block encodings for numerical linear algebraists

How the combination rules are realized
(supplement to "Preconditioning for quantum linear solvers in PDE-constrained optimization")

Martin Stoll John W. Pearson

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Abstract

Section 2 of the paper lists how block encodings combine: products, Kronecker products, linear combinations, stacking, concatenation and sparse matrices, each with its subnormalization. This supplement shows, in pure matrix terms, how each rule is realized as an explicit unitary matrix, and why the subnormalizations are what they are. Every construction was built as a dense matrix and checked numerically, with the script `qp_block_encoding_rules.py` in the folder `preconditioning/code` (Table 2).

1 Conventions

Registers are Kronecker factors. A quantum computer with q qubits acts on $\mathbb{C}^2 \otimes \cdots \otimes \mathbb{C}^2 = \mathbb{C}^{2^q}$. A *register* is a group of these factors. A *circuit* is a product of unitary matrices, each acting on a few factors and as the identity on the rest. Two operations are free:

- permuting tensor factors, which is only a relabelling of qubits;
- adding fresh factors that start in the first basis vector e_0.

Ordering. We write extra registers (ancillas, index registers) as the *outer* Kronecker factors and the system as the innermost one. A vector in $\mathbb{C}^2 \otimes \mathbb{C}^n$ is therefore a stacked vector $[v_0; v_1]$, and $e_0 \otimes v = [v; 0]$.

Block encodings. A unitary $U \in \mathbb{C}^{2^a n \times 2^a n}$ is a block encoding of $A \in \mathbb{C}^{n \times n}$ with subnormalization α if its leading block is A/α:

$$(e_0^T \otimes I_n)\, U\, (e_0 \otimes I_n) = A/\alpha, \qquad \text{i.e.} \qquad U = \begin{bmatrix} A/\alpha & * \\ * & * \end{bmatrix}.$$

Applying U to $[v; 0]$ gives $[Av/\alpha;\ w]$ for some w. Measuring the ancilla and keeping the outcome "0" (*postselection*) leaves $Av/\|Av\|$, with probability $\|Av\|^2/(\alpha\|v\|)^2$.

For rectangular $A \in \mathbb{C}^{m \times n}$ the rows and the columns are selected by different projections. These are P_{out} and P_{in}, both with orthonormal columns, and we require $P_{\text{out}}^* U P_{\text{in}} = A/\alpha$.

Why $\alpha \geq \|A\|$. Every block of a unitary matrix has norm at most one, so $\|A/\alpha\| \leq 1$. The subnormalization is the price of embedding a non-unitary matrix in a unitary one. It enters the cost of everything downstream: success probabilities, and the degree of QSVT polynomials.

2 A single matrix: the unitary dilation

Lemma 1 (Halmos dilation)**.** *Let* $\|X\| \le 1$*. Then*

$$U = \begin{bmatrix} X & (I - XX^*)^{1/2} \\ (I - X^*X)^{1/2} & -X^* \end{bmatrix}$$

is unitary.

Proof. Use $X^* f(XX^*) = f(X^*X)X^*$ for $f(t) = (1-t)^{1/2}$. This makes the off-diagonal blocks of U^*U vanish, and

$$U^*U = \begin{bmatrix} X^*X + (I - X^*X) & 0 \\ 0 & (I - XX^*) + XX^* \end{bmatrix} = I.$$

□

So every A has a block encoding with $\alpha = \|A\|$ and one ancilla qubit. The dilation is dense, however, and has no short circuit in general. The point of the rules below is to build block encodings of *structured* matrices from small, cheap pieces. The price is a subnormalization that may exceed $\|A\|$.

3 The ideal subnormalization and what circuits achieve

Definition. The *ideal subnormalization* of A is $\alpha = \|A\|$. It is a lower bound for every block encoding, and the Halmos dilation attains it. The dilation, however, is dense and has no efficient circuit in general. Efficient block encodings are assembled from structured pieces by the rules of the following sections. At each step the subnormalization can exceed the norm. The table below summarizes when a rule is tight, i.e. when its α equals the norm of the result.

rule	α versus norm	tight when
AB	$\alpha_A \alpha_B \ge \|A\|\|B\| \ge \|AB\|$	A is maximal on the range of B's top singular vector
$A \otimes B$	$\alpha_A \alpha_B$ vs. $\|A \otimes B\| = \|A\|\|B\|$	always (if α_A, α_B are tight)
$\sum_j c_j A_j$	$\sum_j \|c_j\|\alpha_j \ge \|\sum_j c_j A_j\|$	all terms maximal on a common vector, phases aligned
$[A_0; \ldots; A_{k-1}]$	$(\sum_j \alpha_j^2)^{1/2}$ vs. $\|\sum_j A_j^* A_j\|^{1/2}$	all A_j maximal on a common input vector
$[A_0, \ldots, A_{k-1}]$	$(\sum_j \alpha_j^2)^{1/2}$ vs. $\|\sum_j A_j A_j^*\|^{1/2}$	all A_j maximal on a common output vector
sparse A	$a_{\max}\sqrt{s_r s_c}$ (Schur-test bound) vs. $\|A\|$	rarely

The common mechanism. A loss occurs when the pieces attain their norms on *different* vectors. In PDE discretizations the typical conflict is between oscillatory vectors, which maximize stiffness and gradient operators, and smooth vectors, which maximize mass and prolongation operators.

Examples.

- *A tight LCU.* $I - S$ with the cyclic shift S has $\alpha = 2 = \|I - S\|$ for even n: both terms are maximal on the alternating vector, with aligned signs.

- *An LCU that loses a factor* 2. The 2D stiffness matrix $K = K_1 \otimes M_1 + M_1 \otimes K_1$ as an LCU of two Kronecker products has $\alpha = 2\|K_1\|\|M_1\| \to 8$, while $\|K\| \to 4$ (Q_1, $h = 2^{-L}$; the ratio is 1.993, 2.000, 2.000, 2.000 for $L = 4, 6, 8, 10$). $\|K_1\|$ is attained by the most oscillatory sine vector, $\|M_1\|$ by the smoothest, and no single vector makes both terms maximal.

- *A product that loses* $2^{L-\ell}$. $\|F_1\|\|P_\ell\|/\|F_1 P_\ell\|$ (gradient factor times prolongation) equals $2^{L-\ell}$ up to the coarsest levels (paper, Table 4), for the same reason. The commuting lemma of the paper removes this loss by applying the gradient on the coarse level first.

- *Concatenation over levels loses* $\sqrt{L}$. For $G = [T_1 G_1, \ldots, T_L G_L]$ the rule gives $\alpha \approx \sqrt{L} \max_\ell \|G_\ell\|$. Multilevel theory (a strengthened Cauchy–Schwarz inequality between levels) keeps $\|G\| = O(1)$, because different levels mostly act on different frequency bands. Their output ranges are far from aligned, so the rule is not tight.

- *Sparse access loses a factor 3* in the random example of Table 2: $\alpha = 15.3$ against $\|A\| = 5.1$.

What this means for the paper. The preconditioned KKT matrix is encoded as $\mathcal{B} = \mathcal{G}^T \mathcal{J} \mathcal{G}$ with $\mathcal{G} = I_2 \otimes G$, where $G = [F_{\text{val}} R; \beta^{1/4} F_\nabla R]$ maps multilevel coefficients to values and gradients at quadrature points. The reflection $\mathcal{J}$ has $\alpha = 1$, so $\alpha_\mathcal{B} = \alpha_G^2$.

- *Ideal case:* $\alpha_G = \|G\|$, giving $\alpha_\mathcal{B} = \|G\|^2 = \|R^T A_\beta R\| \le c_1$. The paper's Table 1 has $\|G\| \approx 1.7$–2.7, hence $\kappa_\alpha(\mathcal{B}) \approx 5$–$10$.

- *Even the ideal case is above* $\|\mathcal{B}\|$: $\|\mathcal{B}\| \le \|G\|^2$ since $\|\mathcal{J}\| = 1$. This is the price of writing an indefinite matrix with a reflection in the middle. Here both quantities are $O(1)$.

- *Actual construction:* $\|G\| \le \alpha_G \le C_d \sqrt{L}$, with losses from three places: the 1D LCUs, the stack of values and gradients, and the concatenation over levels. The pessimistic sparse-access constant is $C_3 \approx 45$.

- *Why measuring* α_G *matters:* it enters the cost squared. For $d = 3$ the gap between the bounds is up to $(45\sqrt{L}/2)^2 \approx 500L$. Building the encoding explicitly and measuring α_G (to-do item 1.1) decides where between the two bounds we are.

4 Products: fresh ancillas

Let $U_A = \begin{bmatrix} X_A & Y_A \\ Z_A & * \end{bmatrix}$ and $U_B = \begin{bmatrix} X_B & Y_B \\ Z_B & * \end{bmatrix}$ be block encodings with $X_A = A/\alpha_A$ and $X_B = B/\alpha_B$.

The naive product fails. If both use the *same* ancilla,

$$U_A U_B = \begin{bmatrix} X_A X_B + Y_A Z_B & * \\ * & * \end{bmatrix}.$$

The "garbage" $Z_B v$ that U_B moves into the ancilla-1 block is brought back into the leading block by Y_A.

The construction. Give each factor its own ancilla. Order the space as (ancilla of A) $\otimes$ (ancilla of B) $\otimes$ system. Let $\widetilde{U}_A$ act as U_A on the ancilla of A and the system, and as the identity on the ancilla of B; define $\widetilde{U}_B$ in the same way. Then

$$\left(e_0 \otimes e_0 \otimes I\right)^T \widetilde{U}_A \widetilde{U}_B \left(e_0 \otimes e_0 \otimes I\right) = X_A X_B = \frac{AB}{\alpha_A \alpha_B}.$$

The reason is that $\widetilde{U}_B$ puts its garbage into the block where the ancilla of B equals 1. $\widetilde{U}_A$ never touches that ancilla, so the garbage stays orthogonal to the leading block.

- *Cost:* one call to each factor, plus the extra ancillas.
- *Subnormalization:* $\alpha_A \alpha_B$, which can be much larger than $\|AB\|$. This is the mechanism behind the failure of the naive product encoding of $\widetilde{R}^T \mathcal{A} \widetilde{R}$ in the paper.

5 Kronecker products

$U_A \otimes U_B$ acts on (ancilla$_A$, system$_A$, ancilla$_B$, system$_B$). After the free permutation to (ancilla$_A$, ancilla$_B$, system$_A$, system$_B$), its leading block is

$$X_A \otimes X_B = \frac{A \otimes B}{\alpha_A \alpha_B}.$$

No calls are combined at all: the two circuits run side by side on disjoint qubits. This is why the Kronecker structure of Q_1 matrices on Cartesian grids is so valuable. A d-dimensional factor costs d one-dimensional circuits.

6 Linear combinations of unitaries (LCU)

Let $\sum_{j=0}^{k-1} c_j A_j$ be given, with block encodings U_j of A_j (subnormalization α_j), $k = 2^m$ terms, and $c_j > 0$. Complex phases of c_j are absorbed into U_j. Let $s = \sum_j c_j \alpha_j$.

- *PREPARE:* a unitary $V \in \mathbb{C}^{k \times k}$ with first column $Ve_0 = p$, where $p_j = \sqrt{c_j \alpha_j / s}$. It acts on an index register of m qubits.
- *SELECT:* the block-diagonal unitary SEL = blkdiag$(U_0, \ldots, U_{k-1})$, i.e. "apply U_j if the index register is e_j".

Then

$$W = (V^* \otimes I)\,\mathrm{SEL}\,(V \otimes I)$$

has the leading block (index 0, ancilla 0)

$$\sum_j \bar{p}_j\, p_j\, \frac{A_j}{\alpha_j} = \sum_j \frac{c_j \alpha_j}{s} \frac{A_j}{\alpha_j} = \frac{1}{s} \sum_j c_j A_j.$$

So $\alpha = \sum_j c_j \alpha_j$.

Example: the difference matrix. Let S be the cyclic shift, a permutation matrix and hence unitary, with $\alpha = 1$. With $k = 2$, $p = \frac{1}{\sqrt{2}}(1,1)$, $V = H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ and SEL $=$ blkdiag$(I, -S)$,

$$W = (H \otimes I) \begin{bmatrix} I & 0 \\ 0 & -S \end{bmatrix} (H \otimes I) = \frac{1}{2} \begin{bmatrix} I - S & I + S \\ I + S & I - S \end{bmatrix},$$

a block encoding of $I - S$ with $\alpha = 2$. The 1D quadrature factors F_0 and F_1 of the paper are built this way. They are two-term combinations of shifts with diagonal weights.

7 Stacking

The stack $\begin{bmatrix} A_0 \\ \vdots \\ A_{k-1} \end{bmatrix} \in \mathbb{C}^{kn \times n}$ is tall, and the index register becomes part of the *output*. Take V with $Ve_0 = p$, where $p_j = \alpha_j / N$ and $N = \left(\sum_j \alpha_j^2\right)^{1/2}$, and

$$W = \mathrm{SEL}\,(V \otimes I).$$

On an input $e_0 \otimes e_0 \otimes v$ (index 0, ancilla 0), V produces $\sum_j p_j\, e_j \otimes e_0 \otimes v$. SEL then applies U_j in branch j. Keeping ancilla 0 but *all* index values gives

$$\sum_j e_j \otimes \frac{p_j}{\alpha_j} A_j v = \frac{1}{N} \begin{bmatrix} A_0 v \\ \vdots \\ A_{k-1} v \end{bmatrix}.$$

This is a rectangular block encoding with P_{in}: (index 0, ancilla 0), and P_{out}: (any index, ancilla 0). Its subnormalization is $N = \left(\sum_j \alpha_j^2\right)^{1/2}$. In the paper this stacks the value rows and the d gradient rows of G_ℓ.

8 Concatenation

The concatenation $[A_0, \ldots, A_{k-1}] \in \mathbb{C}^{n \times kn}$ is wide, and the index register is part of the *input*. With the same V,

$$W = (V^* \otimes I)\,\mathrm{SEL}.$$

An input $\sum_j e_j \otimes e_0 \otimes v_j$ goes to $\sum_j e_j \otimes (e_0 \otimes A_j v_j / \alpha_j + \ldots)$. Applying V^* and keeping index 0 and ancilla 0 gives

$$\sum_j \bar{p}_j \frac{A_j v_j}{\alpha_j} = \frac{1}{N} \sum_j A_j v_j.$$

So $\alpha = N = \left(\sum_j \alpha_j^2\right)^{1/2} \le \sqrt{k} \max_j \alpha_j$. The factor $\sqrt{k}$ is the *price of summing*: the vectors $A_j v_j$ add up coherently only in the one index branch that is kept. In the paper this is the concatenation over the L levels, $G = [T_1 G_1, \ldots, T_L G_L]$. It is the source of $\alpha_G = O(\sqrt{L})$, and the reason the paper asks whether the $\sqrt{L}$ can be avoided.

9 Sparse matrices

Let $A \in \mathbb{C}^{n \times n}$ have at most s_r nonzeros per row and s_c per column, and let $a_{\max} = \max |a_{ij}|$. Work in $\mathbb{C}^n \otimes \mathbb{C}^n \otimes \mathbb{C}^2$ and define, for each row i and column j, the unit vectors

$$\psi_i = \frac{1}{\sqrt{s_r}} \sum_{j \in \mathcal{R}_i} e_i \otimes e_j \otimes \left(\frac{\bar{a}_{ij}}{a_{\max}} e_0 + \sqrt{1 - \tfrac{|a_{ij}|^2}{a_{\max}^2}}\, e_1 \right), \qquad \phi_j = \frac{1}{\sqrt{s_c}} \sum_{i \in \mathcal{C}_j} e_i \otimes e_j \otimes e_0.$$

Here $\mathcal{R}_i$ ($\mathcal{C}_j$) is the nonzero pattern of row i (column j), padded with zero entries to exactly s_r (s_c) indices.

- The ψ_i are orthonormal, because they differ in the first factor. The ϕ_j are orthonormal, because they differ in the second.
- $\psi_i^* \phi_j = a_{ij} / (a_{\max} \sqrt{s_r s_c})$.

With the isometries $T_L = [\psi_1, \ldots, \psi_n]$ and $T_R = [\phi_1, \ldots, \phi_n]$,

$$\frac{A}{a_{\max}\sqrt{s_r s_c}} = T_L^* T_R.$$

Completing T_L and T_R to unitaries $U_L = [T_L,\, T_L^\perp]$ and $U_R = [T_R,\, T_R^\perp]$ gives the block encoding $W = U_L^* U_R$, with $\alpha = a_{\max}\sqrt{s_r s_c}$.

What "entries computable by a circuit" means. U_R maps e_j to ϕ_j: it must produce the positions of the nonzeros of column j (a column-index oracle). U_L maps e_i to ψ_i: it must produce the positions in row i, and rotate a flag qubit by an angle determined by a_{ij} (a value oracle). For structured matrices, such as stencils on Cartesian grids, these are small arithmetic circuits on indices. For unstructured meshes they need lookup tables, which is the data-access problem noted in the paper.

The constant. $\alpha = a_{\max}\sqrt{s_r s_c}$ is exactly the Schur-test bound for $\|A\|$, so it is never smaller than $\|A\|$, and often noticeably larger. In the random example of Table 2, $\alpha = 15.3$ against $\|A\| = 5.1$. This is the origin of the pessimistic constant C_d in the paper's Proposition 3.7, and of the suggestion to use Kronecker and LCU constructions instead of generic sparse access.

10 Summary of costs

rule	realization	extra qubits	subnormalization
single A (dense)	Halmos dilation	1	$\|A\|$
AB	$\widetilde{U}_A \widetilde{U}_B$, fresh ancillas	$a_A + a_B$	$\alpha_A \alpha_B$
$A \otimes B$	$U_A \otimes U_B$, permuted	$a_A + a_B$	$\alpha_A \alpha_B$
$\sum_j c_j A_j$	$(V^* \otimes I)\,\mathrm{SEL}\,(V \otimes I)$	$\log_2 k + \max a_j$	$\sum_j c_j \alpha_j$
$[A_0; \ldots; A_{k-1}]$	$\mathrm{SEL}\,(V \otimes I)$	$\log_2 k + \max a_j$	$(\sum_j \alpha_j^2)^{1/2}$
$[A_0, \ldots, A_{k-1}]$	$(V^* \otimes I)\,\mathrm{SEL}$	$\log_2 k + \max a_j$	$(\sum_j \alpha_j^2)^{1/2} \le \sqrt{k}\max \alpha_j$
sparse A	$U_L^* U_R$	$\log_2 n + 1$	$a_{\max}\sqrt{s_r s_c}$
isometry T	$[T,\, T^\perp]$	as needed	1

Table 1: The combination rules. SEL needs *controlled* versions of the U_j, which typically cost a small constant factor more gates.

11 How the rules build G in the paper

$G = [T_1 G_1, \ldots, T_L G_L]$ is assembled from the inside out:

1. 1D two-term factors (values and derivatives at Gauss points): LCUs of shifts with diagonal weights, α of the order of their norm.
2. d-dimensional factors: Kronecker products of 1D factors, so the α's multiply. They match the norms because the norms of Kronecker products also multiply.
3. G_ℓ: a stack of the value factor and the d gradient factors, weighted by ω_ℓ and $\beta^{1/4}$, with $\alpha(G_\ell) = (\sum_c \alpha_c^2)^{1/2}$.
4. $T_\ell G_\ell$: a product with an isometry, which costs nothing in α.
5. G: a concatenation over the levels, which costs $\sqrt{L}$.

The only places where α can exceed the norm are step 1 (if the LCU is not tight), step 3 (stacking is tight only if all components are equally saturated) and step 5 (the $\sqrt{L}$). Measuring α_G for the explicit construction (to-do item 1.1) quantifies exactly these three losses.

12 Numerical verification

construction	$\|W^*W - I\|_{\max}$	error of the selected block
single matrix (Halmos)	$1.2 \cdot 10^{-15}$	0
product, fresh ancillas	$1.9 \cdot 10^{-15}$	$8.4 \cdot 10^{-17}$
product, *same* ancilla	–	$7.9 \cdot 10^{-1}$ (fails, as predicted)
Kronecker product	$1.3 \cdot 10^{-15}$	$5.6 \cdot 10^{-17}$
LCU, $k = 4$ complex coefficients	$3.3 \cdot 10^{-15}$	$3.9 \cdot 10^{-16}$
stacking, $k = 4$	$1.3 \cdot 10^{-15}$	$3.9 \cdot 10^{-17}$
concatenation, $k = 4$	$1.6 \cdot 10^{-15}$	$3.9 \cdot 10^{-17}$
sparse, $n = 8$, $s_r = s_c = 4$	$1.7 \cdot 10^{-15}$	0

Table 2: Dense checks of all constructions (`qp_block_encoding_rules.py`, random complex matrices, $n = 4$ unless stated).